

VDA11.2

June 2026

Exercise 2 (Accelerated Volume Rendering, 5 Points)

a) Pre-integration in direct volume rendering generates a two-dimensional lookup table that has to be re-computed when the transfer function changes. However, when the transfer function only changes in an interval $[l, r]$, parts of the lookup table remain unaffected. Denote the axes of the lookup table as f and b , for front and back values of a given integration step. Please specify a condition on l, r, f, b that evaluates to True exactly if the given position in the lookup table is affected. (2P) Also provide a sketch that highlights the affected region of the lookup table. (1P)

As both $[l, r]$, and $[b, f]$ (or $[f, b]$ if $f < b$) are intervals, the intersect if and only if either of the endpoints of the latter is contained in the former. A boolean function that works like this would be:

$$\text{update}(l, r, f, b) = (l \leq f \ \& \ f \leq r) \ || \ ((l \leq b \ \& \ b \leq r))$$

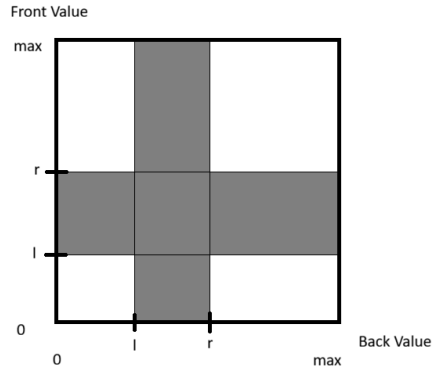


Figure 1: Sketch of the preintegration lookup table with the affected domain marked in grey.

b) Chapter 10, Slide 84 stated the ratio of inner octree nodes to original data points for a full octree in the optimal case of $s \times s \times s$ data points with $s = 2n$. Compute the optimal ratio for 2D quadrees in the same way. (2P)

In the case of quadrees, each step doubles the resolution in each dimension, meaning that in total each level has 4 times the amount of nodes than the previous one. This means that the total amount of nodes is:

$$n_{nodes} = \sum_{i=0}^{\log_2(s)-1} 4^i = \frac{4^{\log_2(s)} - 1}{4 - 1} = \frac{s^2 - 1}{3} \approx \frac{n_{data-points}}{3}$$

Therefore, we deduce that $\frac{n_{nodes}}{n_{data-points}} \approx \frac{1}{3} = 0.333$